Devaj Rathore
2023190

FOB - BIO 101
Assignment 2

1 a) From the graph we can observe that V_{max} is approximately 1.9.

We know that,

$$V = \frac{V_{max}(S)}{K_m + S} \quad \text{(MM equation)}$$

Using values from $S = 4$

$$1.6 = \frac{1.9(4)}{K_m + 4}$$

$$\frac{16}{10}(K_m + 4) = \frac{19}{10}(4)$$

$$4K_m + 16 = 19$$

$$4K_m = 3$$

$$K_m = 0.75$$

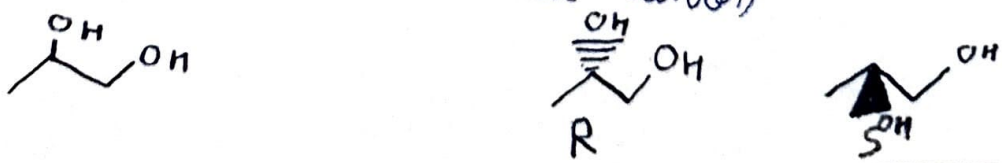
b) When the curve is asymptotic to V_{max}, the reaction order tends to zero.

c) When S is small, the order of reaction tends to 1.

d) this is a hyperbole.

2. The possible structural formulas for $C_3H_8O_2$ are:

- 1,2-propanediol (Has chiral carbon)



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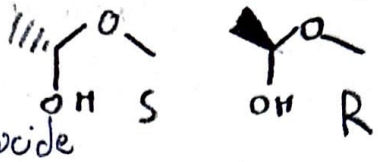
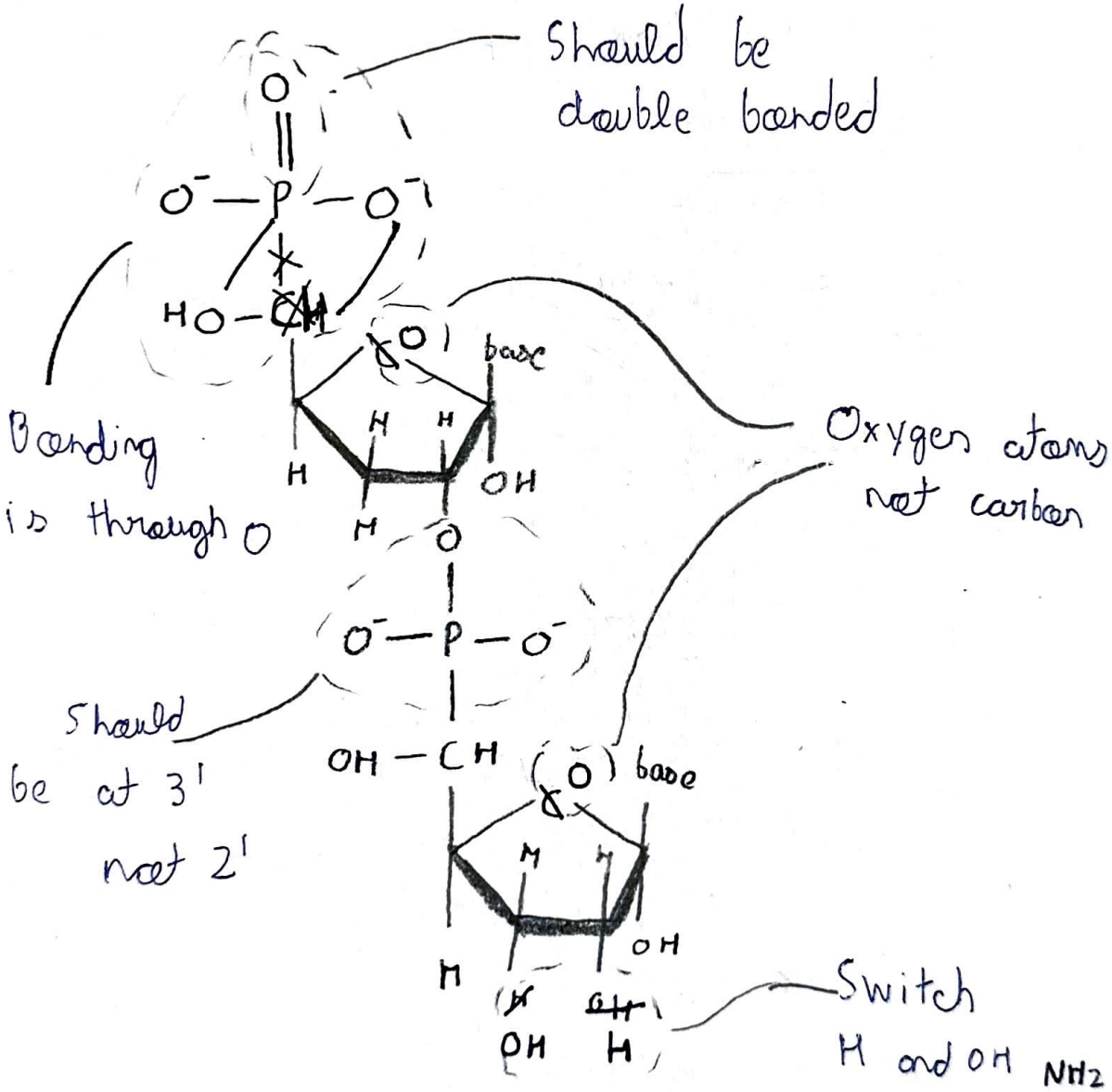
- 1, 3 - Propanedial
- 1, 1- propanedial
- 2, 2- propanedial
- 1-methoxyethanol



2

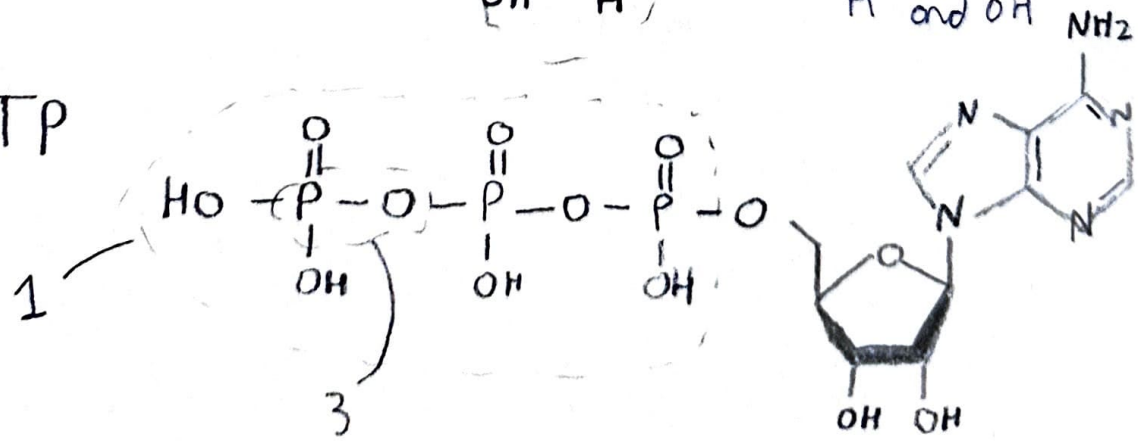
- Propyl hydroperoxide
- 2- hydroperoxy propane
- Ethoxy methanol
- dimethoxy methane
- 2-methoxy methanol

3.



4.

ATP



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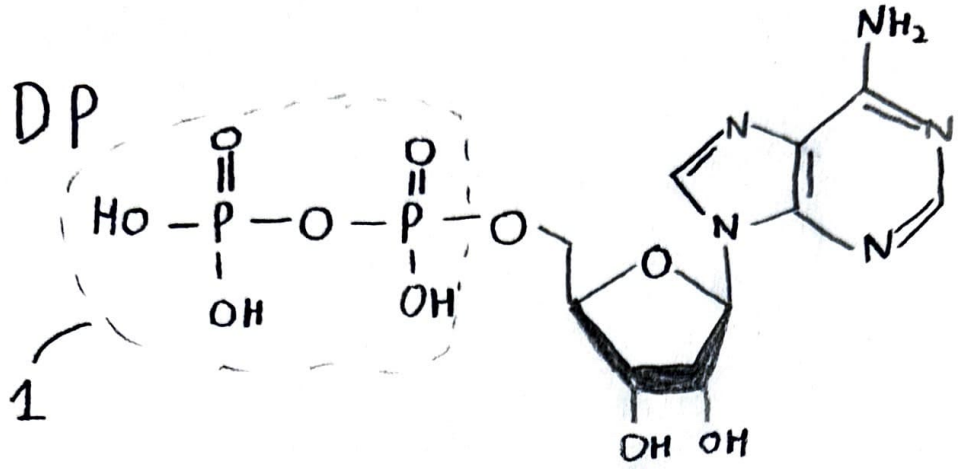
ADP



1. ATP is bulkier resulting in more steric hindrance (3 phosphates) and less stability making it more reactive than ADP (2 phosphates). ~ Steric Hindrance
2. The OHs attatched to phosphate groups increase the negative charge density resulting in electrostatic repulsion. As ATP has more phosphate groups it has higher negative charge density. ~ Electrostatic repulsion.
3. The bond marked has high bond energy, as a result ATP's stability is lowered and its reactivity increases.
4. ATP's additional phosphate group can be hydrolysed easily.
5. In double-stranded DNA molecules :-

Amounts ~~of~~ of :-

Cytosine = Guanine and Adenine = Thymine

Cytosine = 40%. ∴, Guanine = 40%.

Remaining = 20%.

Divided equally b/w Adenine and Thymine.

Adenine = 10%. $\boxed{\text{Thymine = 10\%.}}$